VALEO DESIGN CHALLENGE

(REQUESTED IN 24H)

REAL-TIME REMOTE GPS/LED DEVICE

Solution for the HOME TASK

VALEO Design Challenge Problem Statement

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PRESENTATION PLAN

1. Problem description & main considerations
2. IoT security challenges
3. Recommended IoT software bricks and industrial standards :
    - 3.1 **SESIP™** : An optimized security evaluation methodology, designed for IoT devices
    - 3.2 AWS Greengrass : nearby IoT Hubs
    - 3.3 FreeRTOS : Microcontrollers operating system
    - 3.4 MQTT protocol
    - 3.5 WebSocket protocol
    - 3.6 TensorFlow Lite
4. Secured IoT communication scenarios :
    - 4.1 Switch device LED scenario
    - 4.2 Full scenario with a secured MQTT protocol between the IoT Device, the MQTT Broker and the Mobile
5. Database schema (abstract or simplified)
6. Rest endpoint example
7. Present two architectures (solutions) :
    - 7.1 Custom Solution (based only on open sources)
    - 7.2 Cloud Solution (using AWS)
8. Compare both solutions

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1. REAL-TIME REMOTE GPS/LED DEVICE PROBLEM DESCRIPTION & MAIN CONSIDERATIONS

- The IoT Device must send GPS coordinates to a secured server.
- The IoT Device must run the GPS and communicate efficienlty in term of power consomption.
- An Admin must be able to :
  - Consult in REAL-TIME the fleet of devices
- The IoT device owner must :
  - Consult in REAL-TIME his devices
  - Switch any IoT device LED between ON/OFF from his mobile
- The Machine Learning must run efficienlty on the mobile.

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2. IOT SECURITY CHALLENGES

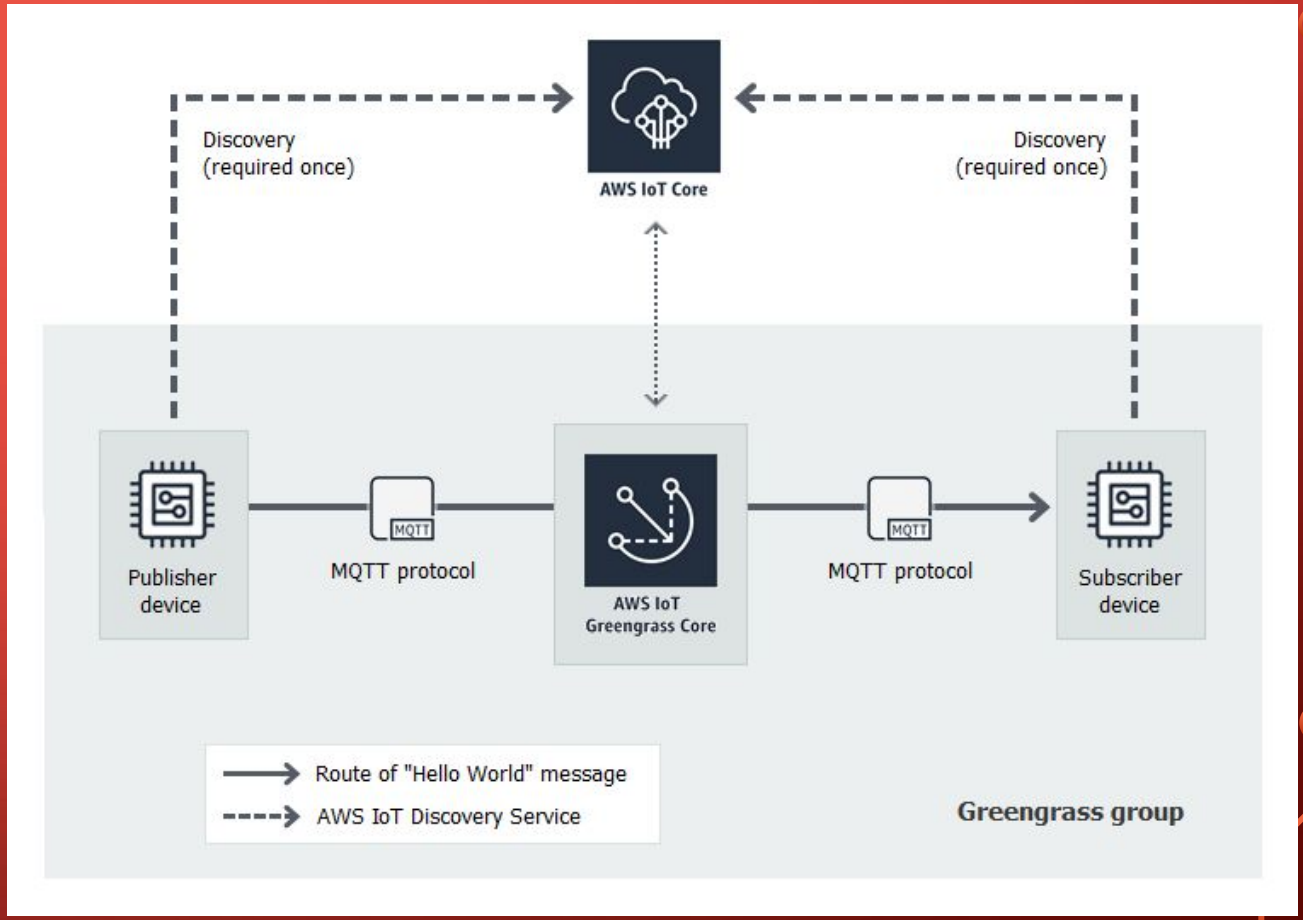
- Weak password protection :
  - In September 2016, MIRAI-infected devices (who became “zombies”) were used to launch the world’s first 1Tbps Distributed Denial-of-Service (DDoS) attack on servers.
  - It took down parts of AWS and its clients, including GitHub, Netflix, Twitter, and Airbnb.
- Lack of regular patches and updates and weak update mechanism
- Lack of insufficient device authentication and authorization and weak or absence of encryption
- Insufficient data protection leads to active eavesdropping attacks (Man-in-The-Middle attack)
- Poor IoT device management :
  - Many medicals health machines were discovered running social media platforms

3.1 SESIP™ STANDARD

- Delivers a flexible and efficient security evaluation methodology dedicated to addressing the complexity of the IoT ecosystem.
- Drives consistency by providing a common and recognized methodology that can be adopted across certification schemes.
- Reduces complexity, cost and time-to-market for IoT stakeholders by offering a methodology that's mappable to other evaluation methodologies, and compliant with standards and regulations.
- Facilitates device certification, by composition of certified parts, and reuse of certification across different evaluations.
- Establishes a consistent and flexible way for IoT developers to demonstrate the security capability of their IoT products and service providers to select a product that matches their security needs.

3.2 AWS Greengrass

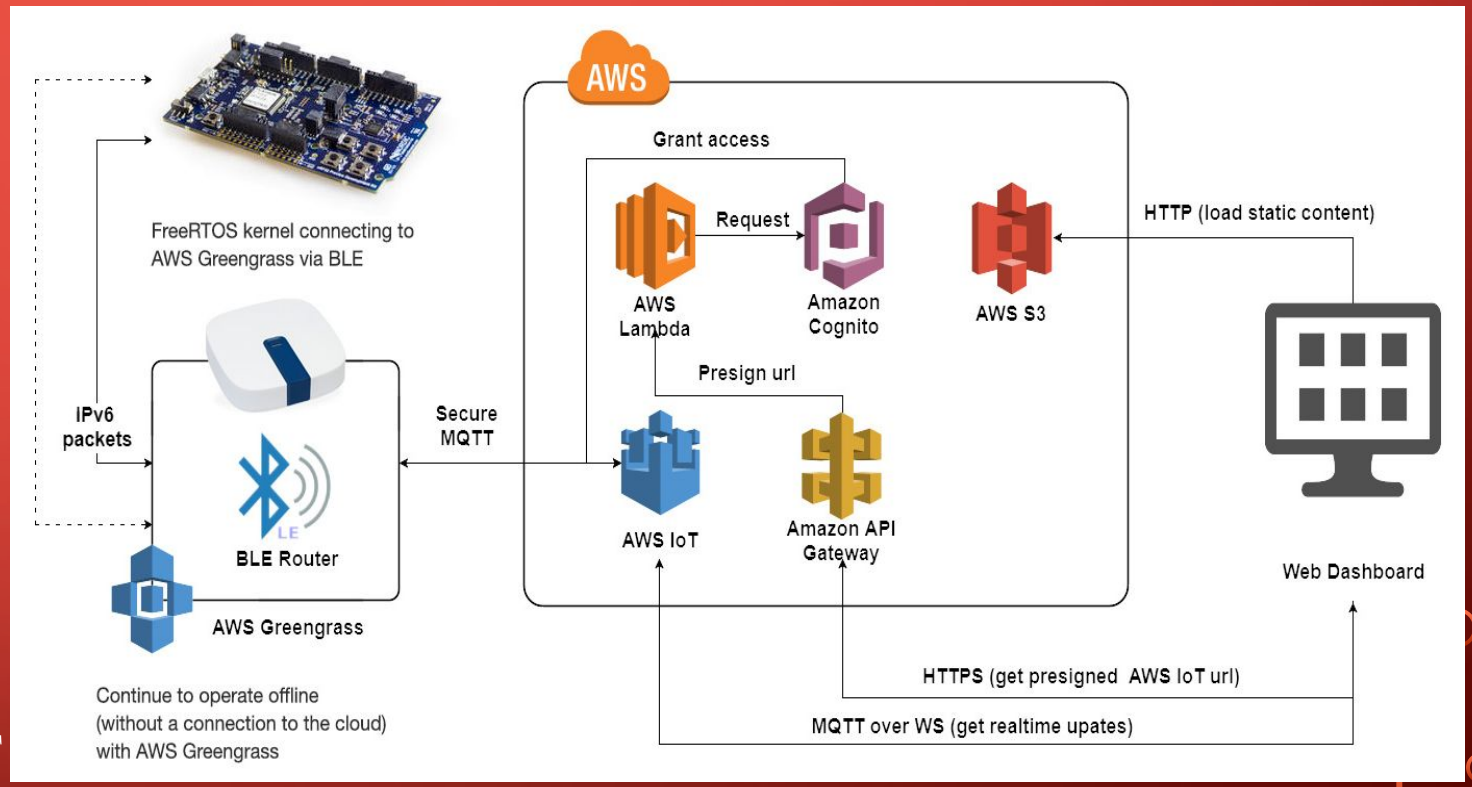
- It’s an IoT open source edge runtime and cloud service that helps to : build, deploy, and manage device software.
- Offer a quick and easy way to build intelligent device software.
- Enables local processing, messaging, data management, ML inference, and offers pre-built components to accelerate application development.
- Enables remote management in the field without needing a firmware update.
- It can act as a hub that communicates with other devices that are running FreeRTOS or using the AWS IoT Device SDK.
- Greengrass is a part of the AWS IoT Device SDK, the Java version 2 supports Android



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3.3 FreeRTOS

- Real-time OS for microcontrollers
- Open source MIT Licence
- Trusted Kernel (standard)
- Supports +40 architectures
- Secured, low-power consumption
- Reduce time to market
- FreeRTOS is now certified for the Security Evaluation Standard for IoT Platforms (**SESIP™**)
- Written in C++



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3.4 MQTT

Subscribing to MQTT channel :

- broker address: eu.valeo.network
- broker port: 1883
- username: gps_tracker_lora
- password:
- topic to subcribe to:

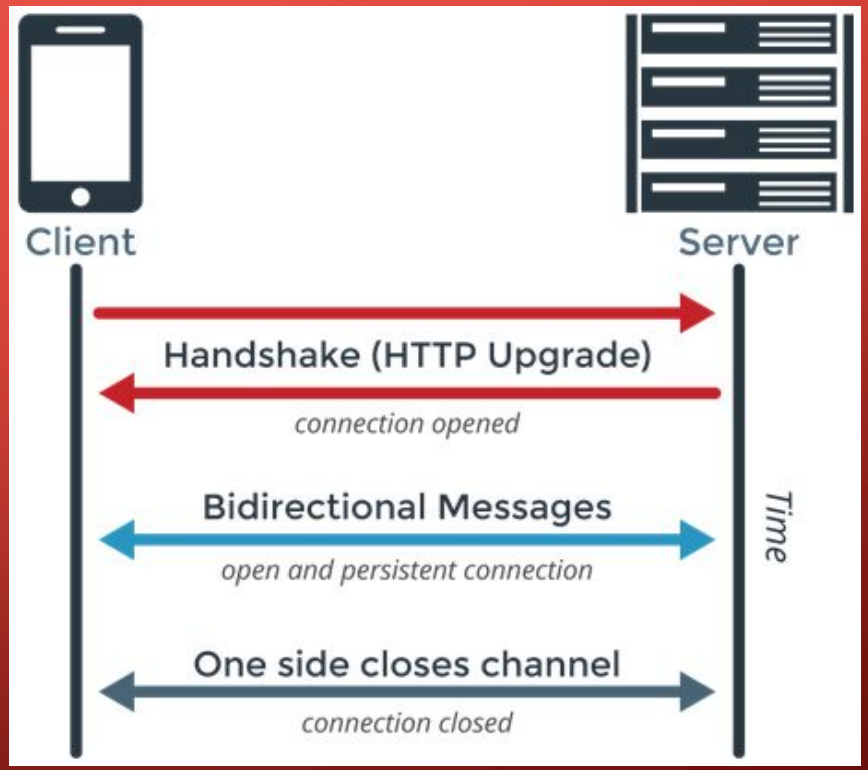
  gps_tracker/devices/<device-id>/up


```
{
"app_id":"gps_tracker_lora","dev_id":"rak811_tracker
","hardware_serial":"00679391F46E8D8E","port":1,"c
ounter":0,"payload_raw":"AYgFN6kQnVsAwPgAFg=
=",
"payload_fields":
{"gps":{"altitude":4.94,"latitude":34.1929,"longitude":
108.8859,"sn":"0 001"}},
      "metadata": {"time":"2019-0725T09:38:32.9569447Z"}
}
```


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3.5 WEBSOCKET PROTOCOL

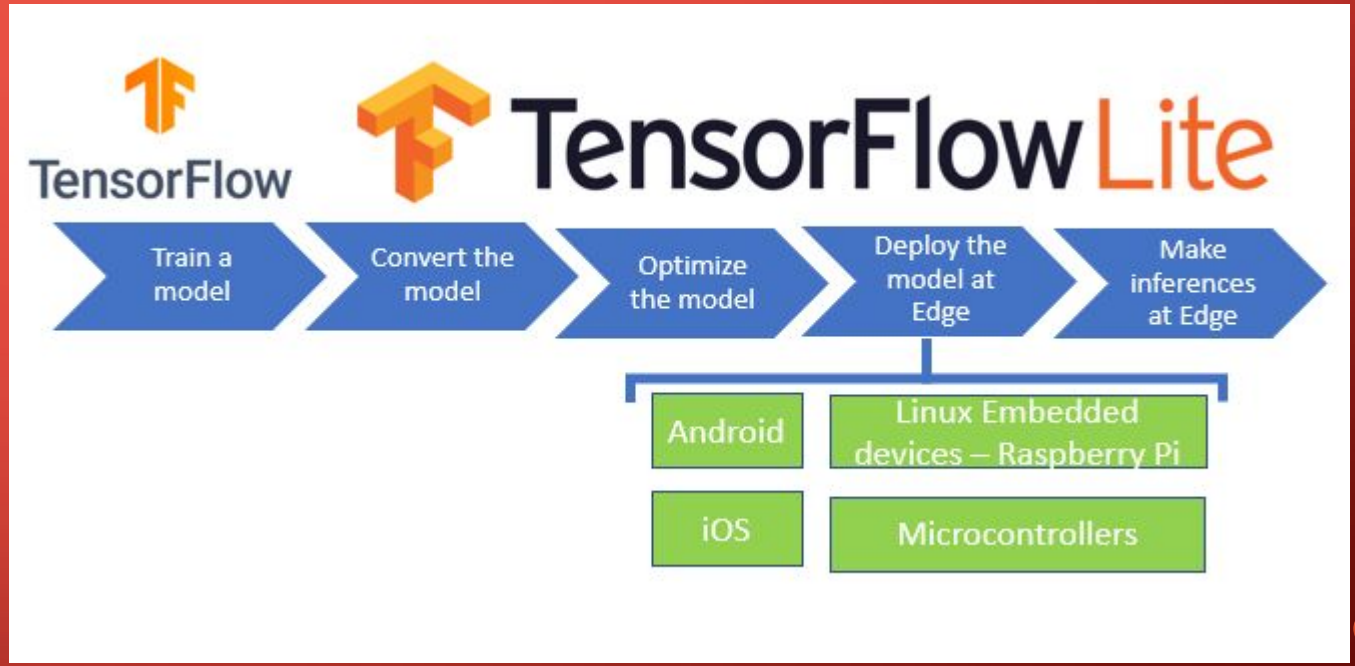
```
var ws = new WebSocket("ws://127.0.0.1:8000/ws/gps/<gps-sn>/");
ws.onmessage = function (msg) {
        # Parse the message
       let json = JSON.parse(msg);
}
```



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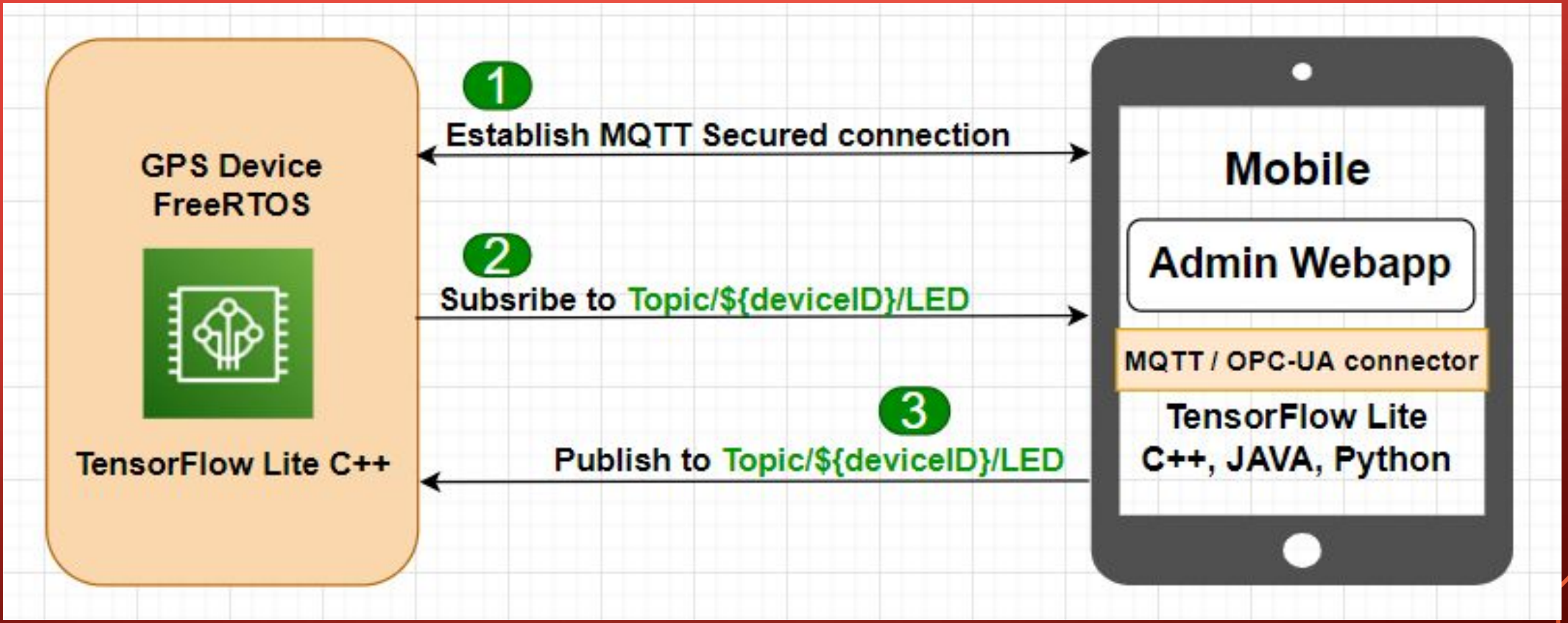
3.6 TensorFlow Lite

- **TensorFlow Lite is an open-source, product ready, cross-platform deep learning framework that converts a pre-trained model in TensorFlow to a special format that can be optimized for speed or storage.**
- **Models are pretrained, light-weight, offer low latency, security, and optimal power consumption (no need for network communication)**
- **Supports edge devices like :**
  - **Mobiles using Android or iOS**
  - **Linux based embedded**
  - **Microcontrollers**

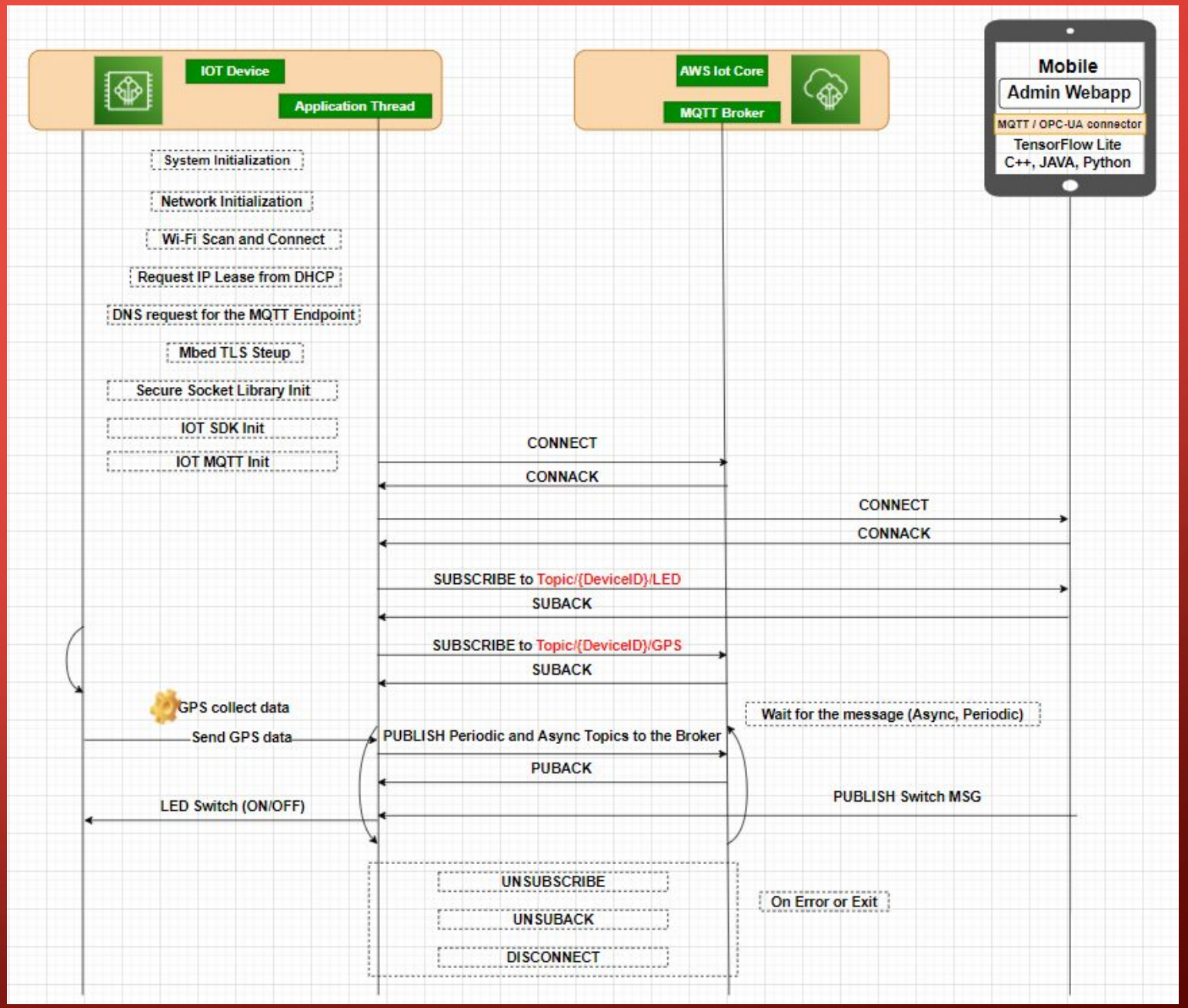


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4.1 Switch device LED scenario

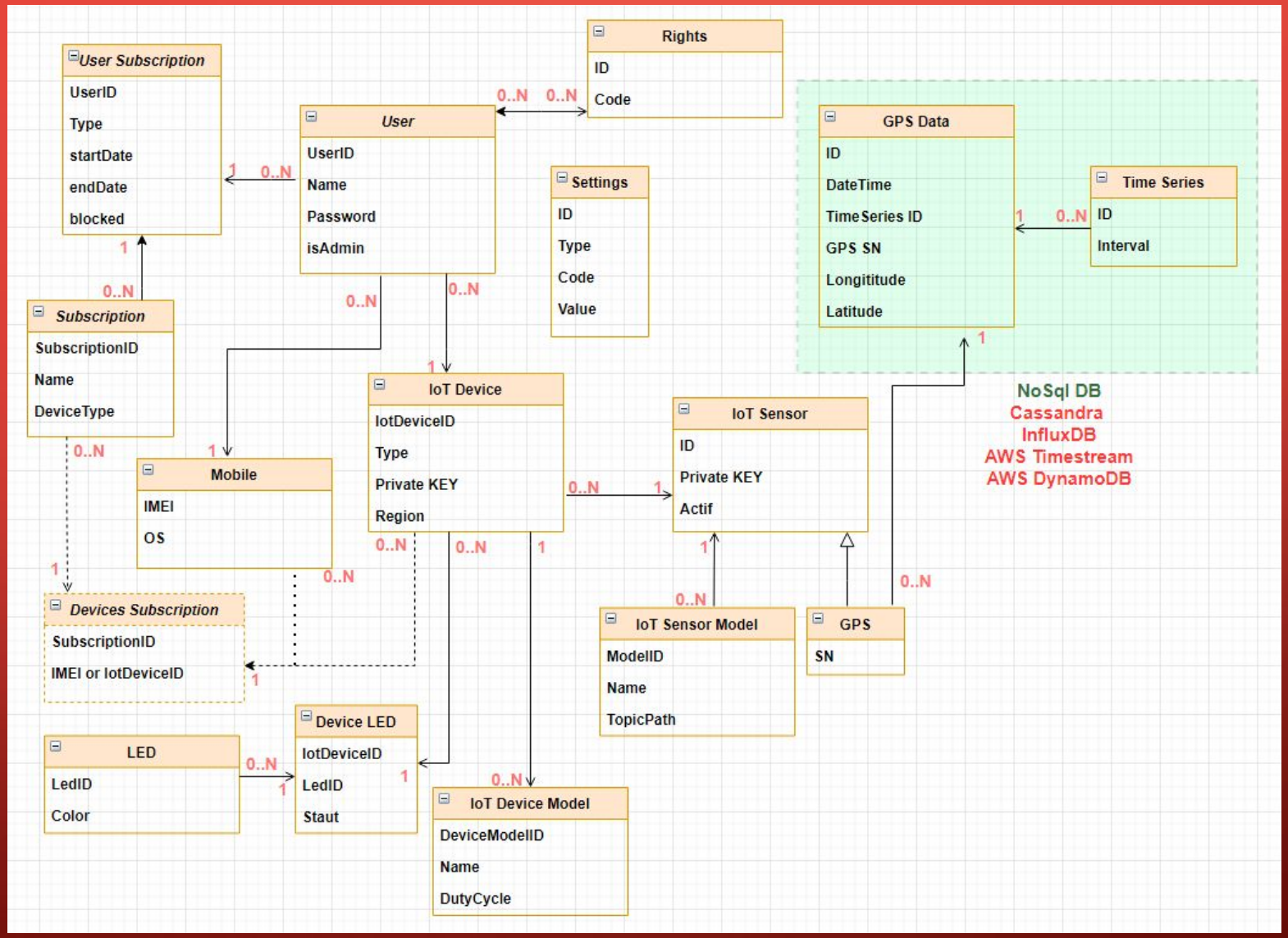


4.2 Full scenario with a secured MQTT protocol between the IoT Device, the MQTT Broker and the mobile

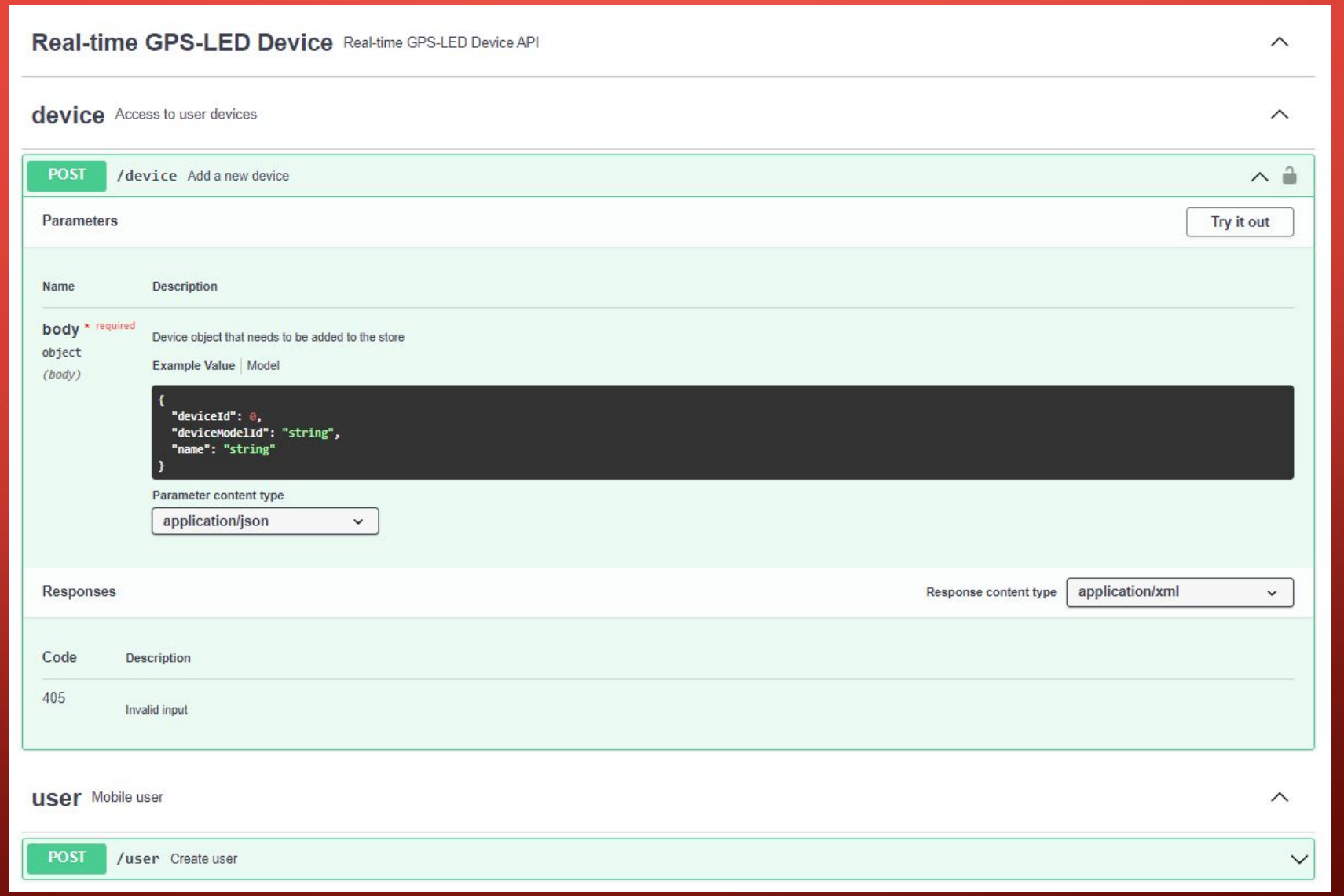


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5. Database schema (abstract or simplified)

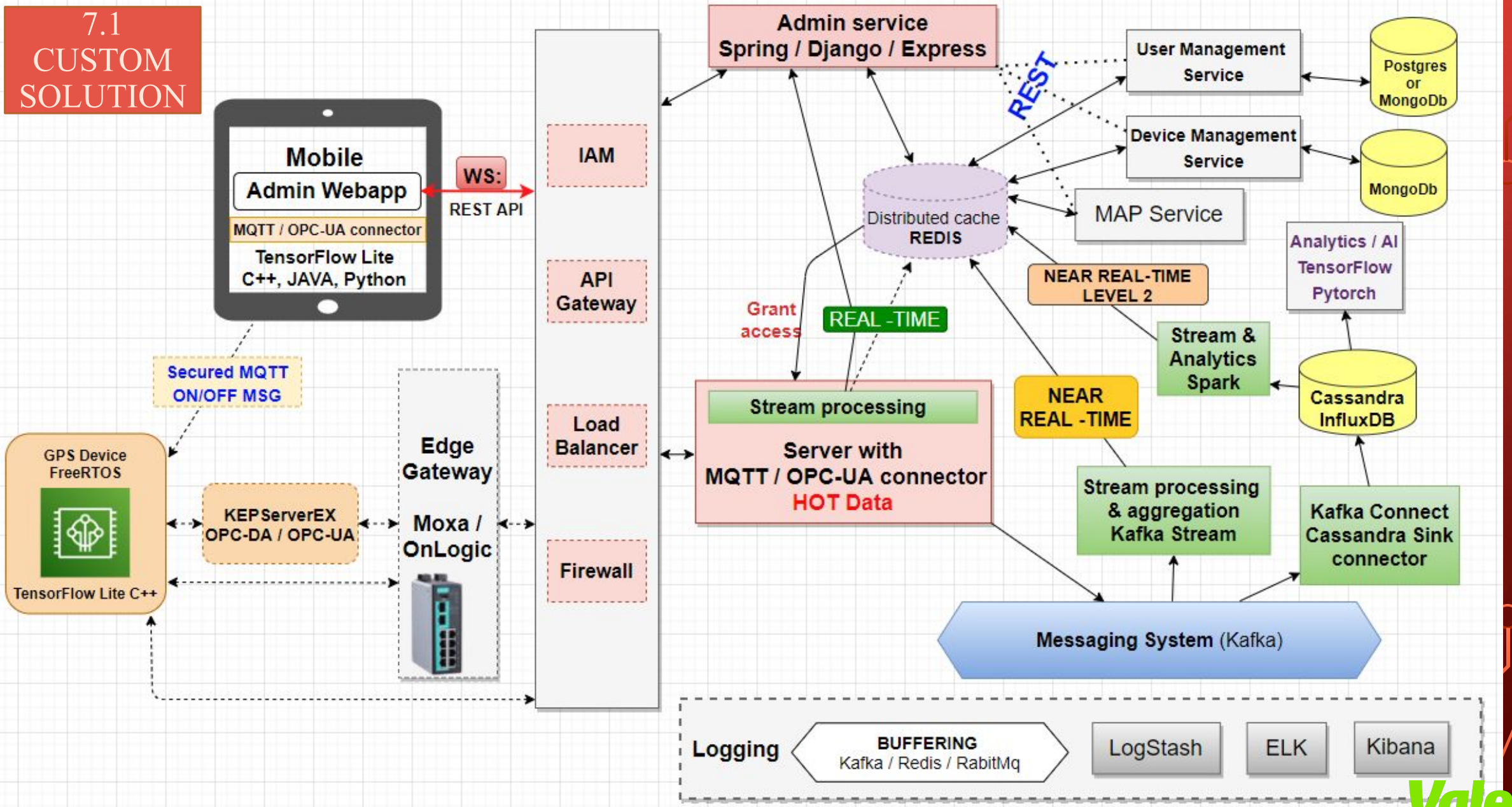


6. Rest end-point example



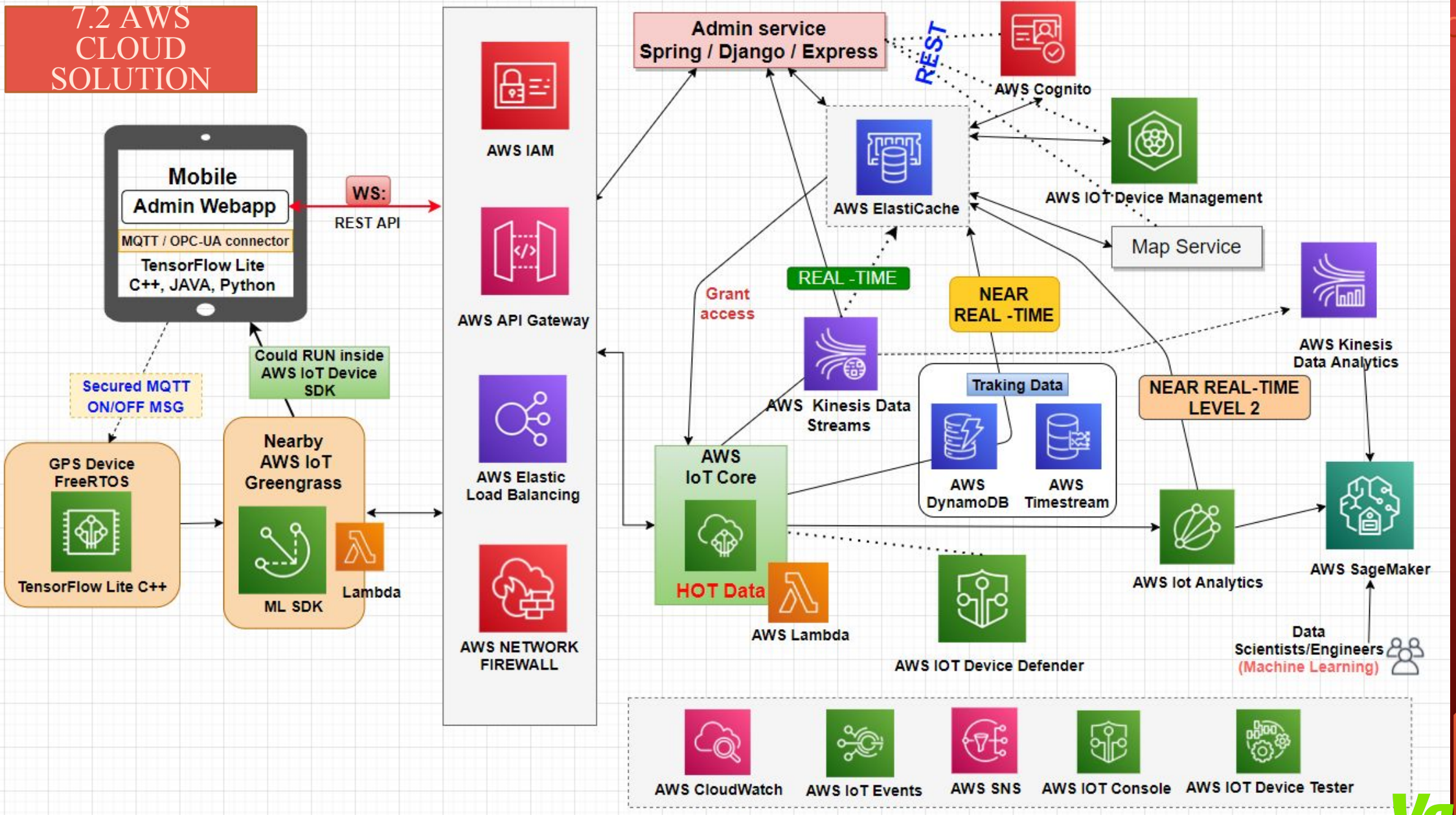
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7.1 CUSTOM SOLUTION



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7.2 AWS CLOUD SOLUTION



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8. CUSTOM VS CLOUD



Aspect	Weight	Custom	Cloud
Development Cost	1	1	0
Exploitation Cost	1	0	1
Maintain & Evolution Cost	1	1	1
Time to market	2	0	1
Security and Resiliency	2	0	1
Developers attraction	1	0	1
Flexibility	2	1	0
Testability	1	1	0
Total		5	8

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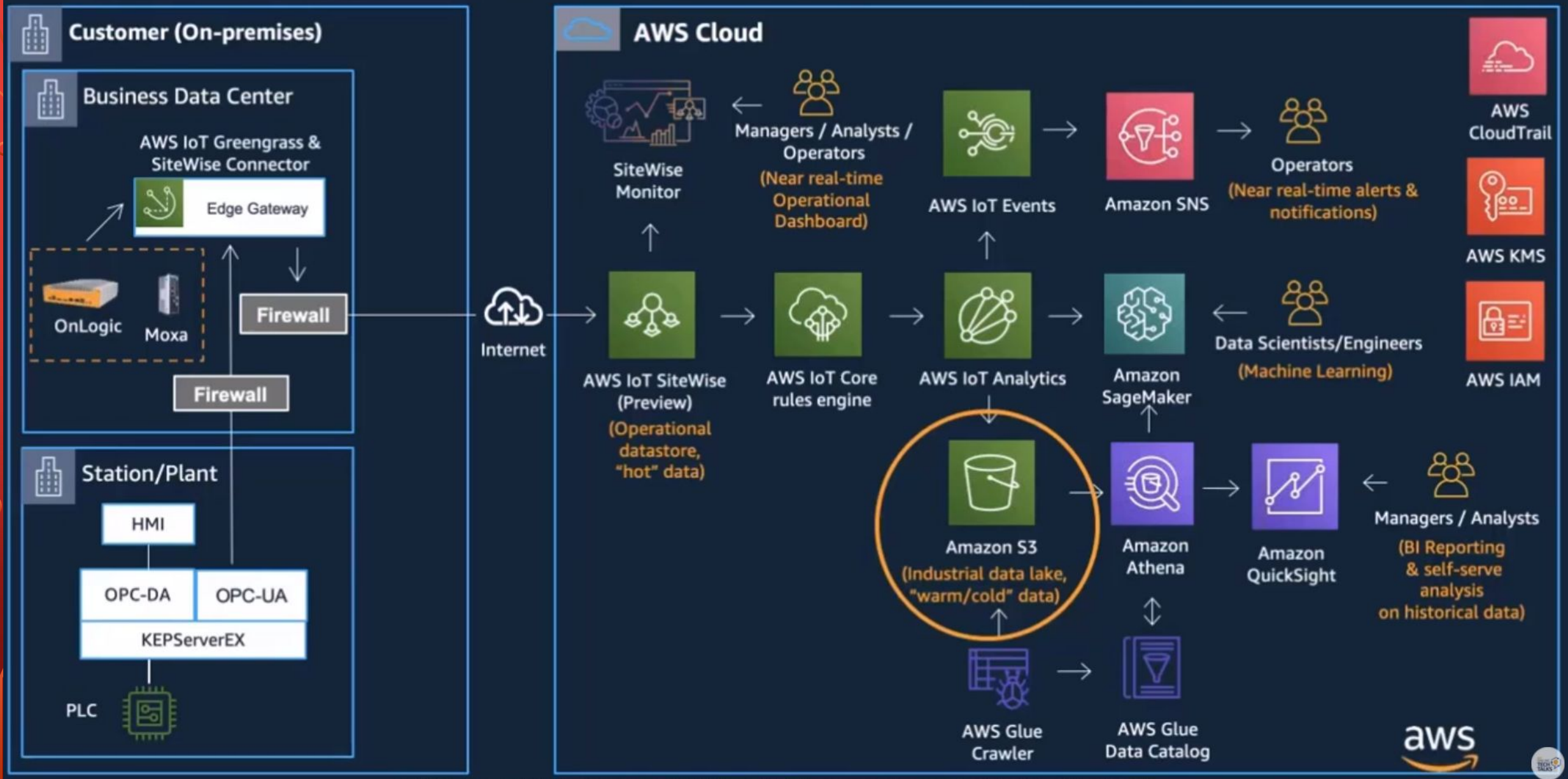
HOW ABOUT QUESTIONS ?



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Core Architecture of a Customer solution



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